AI READINESS ASSESSMENT

Dental School Departments and Courses

Canada · United States · Brazil · China · India

Department/course-specific questionnaire · version 1.0.0 · 2026-10-28

Proposed article title

Course-Level Mapping of Artificial Intelligence Readiness in Dental Education: A Multinational Assessment of Infrastructure, Human Resources, Curriculum Hours and Governance in Canada, the United States, Brazil, China and India

Built on the PAHO/WHO–IDB AI Readiness Assessment Toolkit (2024), AI-GUARD (2026), Bias-free AI in Health (2026) and AI Prompt Design for Public Health (2025).

1. About the study
Objectives
How the questionnaire is applied
Respondent instructions
Country context
Readiness domains and scale
2. Questionnaire
Part A. Named department/course profile (completed by the research team)
Part B. Context of the named department/course
Part C. Physical resources, infrastructure and data
Part D. Human resources and expertise
Part E. Curriculum and teaching initiatives
Part F. Generative AI in teaching and learning
Part G. Governance, ethics, bias and regulation
Part H. Implementation, monitoring, evaluation and sustainability
Part I. Self-assessment and open response
3. Scoring and consolidation
Data file
Methodological recommendations
4. Related studies
Suggested article titles
Candidate journals
References

1. About the study

This cross-sectional study maps how far artificial intelligence (AI) has been integrated into dental education, at the level of named departments and courses rather than whole schools. It covers five of the largest dental-education systems in the world: Canada, the United States, Brazil, China and India.

Published work on AI in dental education is dominated by surveys of students' and educators' perceptions and by single-institution reports. Few studies measure what schools actually provide — computing resources, AI tools used by students, trained faculty, hours of AI instruction by topic (including programming and formal model development), governance and funding. This instrument measures those inputs, course by course, so results can be consolidated to the department, school and country level.

The dimensions and 1–4 maturity scale are adapted from the PAHO/WHO–IDB AI in Public Health Readiness Assessment Toolkit. Governance items for AI tools follow the PAHO AI-GUARD tool (Yes / Partial / No). Bias items follow PAHO's Bias-free AI in Health guidance, and generative-AI items follow PAHO's AI Prompt Design for Public Health.

Objectives

- Describe the physical infrastructure (computing, AI-enabled equipment, software, data, connectivity) available to each named department/course.
- Describe human resources: faculty able to teach AI, AI-focused researchers, training received, technical staff and interdisciplinary partners.
- Quantify AI teaching: dedicated courses at undergraduate and graduate level, hours per term by format and by topic, programming and model-development hours, assessment methods and materials.
- Assess governance, ethics, bias, generative-AI policy and regulatory alignment, using country-specific references.
- Assess implementation, monitoring, evaluation and financial sustainability.
- Consolidate scores into a maturity index per department, school and country, and identify gaps and priorities.

How the questionnaire is applied

- **Step 1.** The research team contacts the department head or course director and completes Part A (profile and consent) with them. Saving Part A on the study website generates a unique 6-character access code (format XXX-XXX).
- **Step 2.** The team sends the code and the website address to the respondent.
- **Step 3.** The respondent enters the code on the website (the only field on the login page) and answers Parts B–I. Answers are saved at every section change, so the form can be completed in several sessions.
- **Step 4.** After submission the form is locked. Each form produces one row in the data file (responses.csv), one column per item, ready for statistical analysis.

Respondent instructions

- Answer every item only for the department/course named in Part A. Do not answer for the school as a whole unless an item says so.
- Where an item asks about a course, write the course code and title.
- If the same department runs more than one AI-related course, a separate form (and code) is issued for each named course.
- Use "Don't know / Not applicable" rather than guessing. Approximate numbers are acceptable.
- Hours are reported per academic term. Item B05 (terms per year) converts them into hours per year; programs with an annual system (e.g., BDS in India) enter 1 term per year.

Country context

Items marked "Country adaptation" are read with the national references below. Option lists that are specific to a country are tagged with the country code (CA, US, BR, CN, IN).

	Canada	United States	Brazil	China	India
Degree	DDS/DMD	DDS/DMD	Bacharelado em Odontologia (Cirurgião-Dentista)	Bachelor of Medicine in Stomatology (口腔医学, 5-year) or integrated 8-year program	BDS (Bachelor of Dental Surgery)
Curriculum / accreditation body	CDAC (Commission on Dental Accreditation of Canada)	CODA (Commission on Dental Accreditation)	MEC/INEP (DCN Odontologia, Res. CNE/CES 3/2021) and CAPES for graduate programs	Ministry of Education (MOE) program accreditation	Dental Council of India (DCI) / National Dental Commission (NDC Act 2023), NAAC
Professional and regulatory bodies	CDA, ACFD and provincial regulatory colleges (e.g., RCDSO, ODQ)	ADA, ADEA and state dental boards	CFO and regional councils (CROs)	Chinese Stomatological Association (CSA) and National Health Commission (NHC)	Indian Dental Association (IDA) and state dental councils
Privacy law	PIPEDA and provincial health-privacy laws (e.g., PHIPA, Quebec Law 25)	HIPAA and FERPA	LGPD (Lei 13.709/2018)	Personal Information Protection Law (PIPL, 2021) and Data Security Law (2021)	Digital Personal Data Protection Act 2023 (DPDP) and its Rules
AI device regulator	Health Canada	FDA	ANVISA	NMPA (National Medical Products Administration)	CDSCO (Medical Devices Rules 2017)
Academic calendar	Most programs use two semesters per year.	Programs use semesters, trimesters or quarters.	Most programs use two semesters per year.	Most programs use two semesters per year.	BDS commonly follows an annual (yearly) examination system; enter 1 term per year in that case.
Languages	English / French	English	Portuguese	Chinese (Mandarin)	English (plus regional languages)

Readiness domains and scale

Maturity items use a 1–4 anchored scale adapted from the PAHO/WHO toolkit (1 = absent, 2 = initial, 3 = developing, 4 = established), plus "Don't know / Not applicable" (coded −1 and treated as missing). Each 1–4 item belongs to one of seven domains:

- **INF —** Physical infrastructure and technology
- **DAT —** Data management, quality and privacy
- **HUM —** Human resources and expertise
- **CUR —** Curriculum and teaching
- **GEN —** Generative AI governance and literacy

- **GOV —** Governance, ethics and regulation
- **IMP —** Implementation, evaluation and sustainability

2. Questionnaire

Part A. Named department/course profile (completed by the research team)

The research team completes this part with the department head or course director (by phone, video call or email). Saving it generates the XXX-XXX access code given to the respondent.

A01. Country. Country *[required]*

- □ Canada
- □ United States
- □ Brazil
- □ China
- □ India

A02. Region. State / province / region *[required]*

Answer: _____

A03. Institution. Name of dental school (institution) *[required]*

Answer: _____

A04. Institution type. Type of institution *[required]*

- □ Public (federal / national)
- □ Public (state / provincial / municipal)
- □ Private, non-profit
- □ Private, for-profit

A05. City. City *[required]*

Answer: _____

A06. Department. Name of the department/course being reported (e.g., Department of Oral Radiology) *[required]*

Answer: _____

A07. Course. Course code and title referenced in this form (e.g., "RAD-301 Oral Radiology II"). Write "Department-wide" if the form covers the whole department. *[required]*

Answer: _____

A08. Level. Level of this named department/course *[required]*

- □ Undergraduate first professional degree (DDS/DMD, BDS, Bacharelado em Odontologia, 口腔医学 Bachelor)
- □ Research master's (MSc/MS, Mestrado acadêmico, 学术型硕士)
- □ Professional/clinical master's or specialty degree (MDS in India, Mestrado profissional, 专业型硕士)
- □ Doctoral (PhD/DSc, Doutorado, 博士)
- □ Residency / specialty training (US/Canada residency, Brazil residência/especialização, China standardized residency training 规培)
- □ Integrated long-track program (e.g., China 8-year program, DDS-PhD)
- □ Continuing education / CPD
- □ Other

Other, specify: _____

A09. Respondent role. Role of the respondent in this named department/course *[required]*

- ☐ Department head / chair
- ☐ Course director / coordinator
- ☐ Program director (graduate / residency)
- ☐ Curriculum committee member
- ☐ Faculty representative
- ☐ Dean / associate dean
- ☐ Other

Other, specify: _____

A10. Respondent name. Respondent full name *[personal data — stored separately]*

Answer: _____

A11. Respondent email. Respondent email *[personal data — stored separately]*

Answer: _____

A12. Interviewer. Research team member who completed Part A *[personal data — stored separately]*

Answer: _____

A13. Language. Preferred language of the respondent

- ☐ English
- ☐ French
- ☐ Portuguese
- ☐ Chinese
- ☐ Hindi
- ☐ Spanish
- ☐ Other

Other, specify: _____

A14. Consent. Informed consent to participate was obtained *[required]*

- ☐ Yes
- ☐ No

Part B. Context of the named department/course

Answer every item only for the department/course named in Part A. Approximate numbers are acceptable.

B01. Enrollment. Approximate number of students enrolled in this department/course per year

Answer (students): _____

B02. Faculty headcount. Number of faculty members assigned to this department/course (headcount)

Answer (faculty): _____

B03. Faculty FTE. Full-time equivalent (FTE) of faculty assigned to this department/course

Answer (FTE): _____

B04. Academic calendar. Academic calendar used by this department/course

Country adaptation: see table "Country context", row "Academic calendar".

- ☐ Semester (2 terms per year)
- ☐ Trimester (3 terms per year)

- ☐ Quarter (4 terms per year)
- ☐ Annual system (1 term per year)
- ☐ Modular / block system

B05. Terms per year. Number of academic terms per year (used to convert hours per term into hours per year)

Answer (terms): _____

B06. Course hours. Total contact hours of this course per term (all topics, not only AI)

Answer (hours per term): _____

B07. Program years. In which program year(s) is this course taught?

- ☐ Year 1
- ☐ Year 2
- ☐ Year 3
- ☐ Year 4
- ☐ Year 5
- ☐ Year 6 or later / internship
- ☐ Graduate / residency year

B08. Course format. Main format of this course

- ☐ Didactic (lectures/seminars)
- ☐ Preclinical / laboratory
- ☐ Clinical
- ☐ Mixed
- ☐ Research

B09. AI courses in department. How many courses in this department include any AI content?

Answer (courses): _____

B10. Respondent AI experience. Your own experience with AI in teaching, research or practice

- ☐ None
- ☐ Less than 1 year
- ☐ 1–3 years
- ☐ More than 3 years

B11. Respondent AI literacy. How would you rate your own AI literacy?

- ☐ (1) Basic: I know general concepts only
- ☐ (2) Intermediate: I can interpret AI outputs and their limitations
- ☐ (3) Advanced: I can critically appraise AI studies and select tools
- ☐ (4) Expert: I develop, train or validate AI models

Part C. Physical resources, infrastructure and data

Facilities, equipment, software, data and technical support available to the named department/course.

C01. Computing hardware. Is there hardware designated for AI-related teaching or research (GPU workstations, high-performance computing, cloud credits) in this department/course? *[domain INF]*

- ☐ (1) None

- ☐ (2) Only individual faculty members' personal equipment
- ☐ (3) Shared equipment, insufficient for demand
- ☐ (4) Dedicated high-performance computing, regularly updated and accessible to students
- ☐ (-1) Don't know / Not applicable

C02. GPU workstations. Number of GPU-equipped workstations or servers available to this department/course

Answer (units): _____

C03. Cloud computing. Access to cloud computing for AI (institutional cloud accounts, research credits, university HPC cluster) *[domain INF]*

- ☐ (1) No access
- ☐ (2) Individual/ad hoc access paid by faculty
- ☐ (3) Institutional access on request, limited quota
- ☐ (4) Routine institutional access with support
- ☐ (-1) Don't know / Not applicable

C04. Student devices ratio. Number of students per institutionally provided computer during hands-on AI or digital sessions (e.g., 1 = one computer per student)

Answer (students per computer): _____

C05. Dedicated digital/AI space. Is there a dedicated physical space for digital dentistry or AI teaching/research (lab, innovation hub)? *[domain INF]*

- ☐ (1) None
- ☐ (2) Shared general computer room
- ☐ (3) Dedicated digital dentistry lab without AI focus
- ☐ (4) Dedicated AI/digital dentistry lab with staff and scheduled student access
- ☐ (-1) Don't know / Not applicable

C06. Software licences. Which AI/data-science software is available to students and faculty of this department/course?

- ☐ Python / Jupyter / Google Colab
- ☐ R / RStudio
- ☐ MATLAB
- ☐ No-code machine learning (e.g., Orange, KNIME, Teachable Machine)
- ☐ Image annotation tools (e.g., CVAT, Label Studio, 3D Slicer)
- ☐ Statistical packages (SPSS, Stata, SAS)
- ☐ Commercial dental AI platforms
- ☐ Institutionally licensed generative AI (e.g., Copilot, ChatGPT Edu, Gemini)
- ☐ None

Other, specify: _____

C07. AI-enabled equipment. Which AI-enabled equipment or systems are in use in this department/course?

- ☐ Digital radiography with AI analysis
- ☐ CBCT with AI segmentation
- ☐ Intraoral scanners with AI features
- ☐ CAD/CAM or 3D printing with AI design

- ☐ Haptic / simulation units with AI feedback
- ☐ Electronic health record with AI modules
- ☐ Learning management system with AI features
- ☐ AI tutor / chatbot for students
- ☐ None

Other, specify: _____

C08. Clinical/educational AI tools. List each AI tool students actually use in this department/course. Leave rows empty if not applicable.

#	Tool / product name	Main task	Regulatory status	Student use	Part of graded work?	Year introduced
1						
2						
3						
4						
5						
6						

Main task codes: caries = Caries detection on radiographs; perio = Periodontal bone-loss assessment; periapical = Periapical lesion detection; cbct = CBCT segmentation / 3D analysis; ceph = Cephalometric landmarking / orthodontic analysis; implant = Implant planning / guided surgery; cad = CAD/CAM restoration design; ios = Intraoral-scan analysis / smile design; lesion = Oral mucosal lesion / oral cancer screening; charting = Charting, clinical notes, ambient scribe; patient = Patient communication / education; admin = Scheduling, billing, administration; sim = Preclinical simulation feedback / preparation grading; llm = General-purpose LLM / chatbot; other = Other

Regulatory status codes: 1 = Cleared/approved by the national regulator for this use; 2 = Foreign approval only (e.g., CE, FDA) ; 3 = Commercial, not cleared; 4 = Research / in-house prototype; -1 = Don't know

Student use codes: 1 = Students observe faculty use; 2 = Preclinical / simulation only; 3 = Students operate under supervision on patients; 4 = Students operate independently in patient care

Part of graded work? codes: 1 = Yes; 0 = No

C09. Number of AI tools. Overall use of AI tools by students in this department/course *[domain INF]*

- ☐ (1) No AI tool is used
- ☐ (2) 1 AI tool used
- ☐ (3) 2–3 AI tools used
- ☐ (4) 4 or more AI tools used and integrated into graded coursework
- ☐ (-1) Don't know / Not applicable

C10. EHR / imaging systems. Name the electronic health record and imaging (PACS) systems used in this department/course (e.g., axiUm, Dentrix, MiPACS)

Answer: _____

C11. De-identified dataset. Is there a standardized, de-identified oral-health dataset (radiographs, intraoral scans, treatment records) that students or faculty of this department/course can access for AI training or research? *[domain DAT]*

- ☐ (1) No dataset accessible
- ☐ (2) Dataset in planning or pilot stage
- ☐ (3) Dataset exists but access is restricted and scale is limited
- ☐ (4) Dataset exists with clear access protocols and is used in the course

- ☐ (-1) Don't know / Not applicable

C12. Dataset modalities. Which data modalities are available in that dataset?

- ☐ Periapical / bitewing radiographs
- ☐ Panoramic radiographs
- ☐ Cephalometric radiographs
- ☐ CBCT
- ☐ Intraoral scans (3D)
- ☐ Clinical photographs
- ☐ Histopathology images
- ☐ Structured treatment / EHR records
- ☐ Free-text clinical notes
- ☐ No dataset

C13. Dataset size. Approximate number of de-identified images or records in that dataset (enter 0 if none)

Answer (images/records): _____

C14. Annotation capacity. Capacity to label/annotate data for AI (calibrated annotators, annotation protocol, inter-rater agreement) *[domain DAT]*

- ☐ (1) No capacity
- ☐ (2) Ad hoc annotation by individual researchers
- ☐ (3) Annotation protocol exists, limited calibration
- ☐ (4) Calibrated annotators with documented protocol and agreement statistics
- ☐ (-1) Don't know / Not applicable

C15. Data quality. What measures ensure the accuracy, completeness and timeliness of data used in this department/course? *[domain DAT]*

- ☐ (1) No measures applied
- ☐ (2) Ad hoc checks only
- ☐ (3) Some measures exist but are not comprehensive
- ☐ (4) Systematic quality assurance with documented protocols
- ☐ (-1) Don't know / Not applicable

C16. Data representativeness. Is the representativeness of the data assessed (age, sex, ethnicity, socioeconomic group, rural/urban, device type)? *[domain DAT]*

- ☐ (1) Never assessed
- ☐ (2) Discussed informally
- ☐ (3) Assessed for some projects
- ☐ (4) Systematically assessed and reported
- ☐ (-1) Don't know / Not applicable

C17. Privacy and security. What privacy and security systems protect patient data used in this department/course? *[domain DAT]*

Country adaptation: see table "Country context", row "Privacy law".

- ☐ (1) No systems
- ☐ (2) Basic password protection only
- ☐ (3) Systems exist but are inconsistent
- ☐ (4) Comprehensive systems with encryption, audit trails and a named person responsible

- ☐ (-1) Don't know / Not applicable

C18. De-identification. Are there documented procedures for anonymization / de-identification of patient data before use in teaching or AI research? *[domain DAT]*

- ☐ (1) No procedures
- ☐ (2) Informal practice only
- ☐ (3) Written procedures, inconsistently applied
- ☐ (4) Written procedures, routinely applied and audited
- ☐ (-1) Don't know / Not applicable

C19. Secure research environment. Is a secure environment available for working with identifiable or sensitive data (trusted research environment, approved server, data enclave)? *[domain DAT]*

- ☐ (1) Not available
- ☐ (2) Planned
- ☐ (3) Available on request with delays
- ☐ (4) Available and routinely used
- ☐ (-1) Don't know / Not applicable

C20. Connectivity and IT support. Does this department/course have reliable broadband and technical support for the AI tools it uses? *[domain INF]*

- ☐ (1) Neither reliable connectivity nor AI-specific support
- ☐ (2) Basic IT support exists but does not cover AI tools
- ☐ (3) Partial support (software installation only)
- ☐ (4) Dedicated AI / educational-technology support assigned to the department/course
- ☐ (-1) Don't know / Not applicable

C21. Interoperability. Can the AI tools used exchange data with existing systems (EHR, imaging/PACS, learning management system) using standards such as DICOM, HL7 FHIR or LTI? *[domain INF]*

- ☐ (1) No interoperability
- ☐ (2) Some standards under development
- ☐ (3) Standards exist but implementation is inconsistent
- ☐ (4) Well-established standards enabling seamless integration
- ☐ (-1) Don't know / Not applicable

C22. Cybersecurity. Are cybersecurity measures applied to AI tools and data (risk assessment, access control, vendor security review)? *[domain INF]*

- ☐ (1) None
- ☐ (2) Basic institutional measures only
- ☐ (3) Measures exist but not specific to AI tools
- ☐ (4) AI-specific security review before adoption and periodic audit
- ☐ (-1) Don't know / Not applicable

C23. Equitable student access. Do all students in this course have equal access to the AI tools and computing needed (including outside class hours)? *[domain INF]*

- ☐ (1) No — access depends on personal devices/subscriptions
- ☐ (2) Limited institutional access
- ☐ (3) Access during scheduled sessions only
- ☐ (4) Equal access during and outside class hours

- ☐ (-1) Don't know / Not applicable

Part D. Human resources and expertise

Faculty and staff assigned to the named department/course.

D01. Faculty able to teach AI. What proportion of faculty can independently teach AI content (machine-learning principles, interpreting AI outputs, critical appraisal of AI literature)? *[domain HUM]*

- ☐ (1) 0%
- ☐ (2) 1–10%
- ☐ (3) 11–25%
- ☐ (4) More than 25%
- ☐ (-1) Don't know / Not applicable

D02. Faculty teaching AI (n). Number of faculty who currently teach any AI content in this department/course

Answer (faculty): _____

D03. Faculty with formal AI degree. Number of faculty with a formal qualification in AI, computer science, data science or biomedical informatics (degree or university certificate)

Answer (faculty): _____

D04. AI-primary researchers. How many faculty have AI in dentistry or dental education as their primary research area? *[domain HUM]*

- ☐ (1) None
- ☐ (2) 1
- ☐ (3) 2–3
- ☐ (4) 4 or more
- ☐ (-1) Don't know / Not applicable

D05. AI publications (3 y). Number of peer-reviewed AI-related publications by faculty of this department/course in the past 3 years

Answer (publications): _____

D06. AI graduate trainees. Number of graduate students, residents or postdocs currently working on AI projects in this department/course

Answer (trainees): _____

D07. Faculty AI training. What formal AI training have faculty received in the past 24 months? *[domain HUM]*

- ☐ (1) No training received
- ☐ (2) Voluntary, sporadic sessions only
- ☐ (3) Organized training, but low or inconsistent attendance
- ☐ (4) Systematic training with institutional incentives and tracking
- ☐ (-1) Don't know / Not applicable

D08. Training hours per faculty. Average hours of AI training per faculty member in the past 12 months

Answer (hours): _____

D09. Training formats. Formats of AI training available to faculty

- ☐ Institutional workshops / seminars
- ☐ Online courses (MOOCs, e.g., Coursera, edX)

- ☐ University certificate programs
- ☐ Conference sessions (IADR, ADEA, ADEE, SBPqO, IDA, CSA)
- ☐ Peer mentoring / communities of practice
- ☐ Vendor training
- ☐ Sponsored degree programs
- ☐ None

Other, specify: _____

D10. Interdisciplinary partners. Name each external unit this department/course collaborates with on AI teaching or research (e.g., Department of Computer Science, Faculty of Engineering, a company).

#	Partner unit	Type
1		
2		
3		

Type codes: 1 = Same university — computer science / engineering; 2 = Same university — medicine / public health / other; 3 = Other university (national); 4 = Other university (international); 5 = Industry / company; 6 = Government / hospital / other

D11. Partnership depth. Depth of interdisciplinary collaboration on AI *[domain HUM]*

- ☐ (1) No partners
- ☐ (2) Informal individual collaboration only
- ☐ (3) Formal project-based collaboration
- ☐ (4) Long-term institutional partnership with joint funding or appointments
- ☐ (-1) Don't know / Not applicable

D12. Technical staff. Access to dedicated technical personnel (data scientists, AI engineers, statisticians, research software engineers) *[domain HUM]*

- ☐ (1) No access
- ☐ (2) Shared across the school, high demand
- ☐ (3) Adequate access for most needs
- ☐ (4) Dedicated technical staff embedded in the department/course
- ☐ (-1) Don't know / Not applicable

D13. Technical staff FTE. Full-time equivalent of technical AI/data staff available to this department/course

Answer (FTE): _____

D14. AI hiring. Is recruitment of faculty or staff with AI expertise planned? *[domain HUM]*

- ☐ (1) Not planned
- ☐ (2) Discussed, no approved position
- ☐ (3) Position approved or recruitment underway
- ☐ (4) AI expertise already recruited and further positions planned
- ☐ (-1) Don't know / Not applicable

D15. Faculty openness. Overall attitude of faculty in this department/course toward integrating AI *[domain HUM]*

- ☐ (1) Mostly resistant
- ☐ (2) Mixed / indifferent

- ☐ (3) Mostly supportive
- ☐ (4) Strongly supportive with active champions
- ☐ (-1) Don't know / Not applicable

D16. Faculty competency barriers. Most significant barriers to faculty AI competency (select up to two)

Select up to 2.

- ☐ Lack of faculty time
- ☐ Lack of funding for training
- ☐ Lack of faculty interest
- ☐ Lack of qualified trainers
- ☐ Competing curricular priorities
- ☐ No recognition in promotion/tenure
- ☐ Training not available in local language

Other, specify: _____

Part E. Curriculum and teaching initiatives

Refers exclusively to the named course (Part A, item A07). Hours: include lectures, hands-on sessions and assignments; exclude generic literature searching and routine use of clinical software.

E01. Dedicated AI instruction. Does the course include dedicated AI instruction (not a passing mention)? *[domain CUR]*

- ☐ (1) None
- ☐ (2) Mentioned in 1–2 lectures only
- ☐ (3) 1 dedicated module or class session (≤4 hours)
- ☐ (4) 1 or more dedicated AI courses, or systematic integration across sessions
- ☐ (-1) Don't know / Not applicable

E02. AI hours by format. Hours of AI-specific instruction a student receives in this course per term, by format (enter 0 if none)

Topic / format	Hours (hours per term)
Lectures / seminars	
Hands-on / computer lab	
Clinical / preclinical use with supervision	
Assignments / guided self-study	

E03. AI hours by topic. Hours per term devoted to each AI topic in this course (enter 0 if not covered; a session may count toward more than one topic)

Topic / format	Hours (hours per term)
AI fundamentals and terminology	
Machine learning / deep learning principles	
Data science and statistics for AI	
Programming (Python, R)	
Model development and training	

Topic / format	Hours (hours per term)
Model validation and performance metrics (sensitivity, AUC, calibration)	
Bias, fairness and subgroup performance	
Explainability / interpretability	
Clinical interpretation of AI outputs and AI errors	
AI in dental imaging (radiographs, CBCT)	
AI in treatment planning, CAD/CAM, orthodontics, implants	
Generative AI, large language models and prompt design	
Ethics, privacy, informed consent, accountability	
Regulation, standards and approval of AI devices	
Critical appraisal of AI literature and reporting guidelines	
AI as a research tool (literature, analysis, writing)	

E04. Required or elective. Is the AI content of this course required or elective?

- ☐ No AI content
- ☐ Elective / optional
- ☐ Required for some students / tracks
- ☐ Required for all students

E05. AI credits. Academic credits awarded for AI content (local credit units; 0 if none)

Answer (credits): _____

E06. Technical depth. Which best describes the technical depth of AI instruction? *[domain CUR]*

- ☐ (1) Conceptual awareness only
- ☐ (2) Clinical interpretation of AI outputs
- ☐ (3) Hands-on operation of existing AI tools under guidance
- ☐ (4) Includes programming or formal AI development
- ☐ (-1) Don't know / Not applicable

E07. Programming / AI development. Does the course cover programming (Python/R) or formal AI development (model training, validation, bias assessment)? *[domain CUR]*

- ☐ (1) None
- ☐ (2) Instructor demonstration only
- ☐ (3) Hands-on exercises under guidance
- ☐ (4) Students independently complete a small AI project (e.g., train and validate a classifier)
- ☐ (-1) Don't know / Not applicable

E08. Development platforms. Platforms used for programming or model development

- ☐ Python / Jupyter / Colab
- ☐ R
- ☐ No-code tool (Orange, KNIME, Teachable Machine)
- ☐ Cloud AutoML (Google, Azure, AWS)
- ☐ Vendor research platform

□ Not applicable

Other, specify: _____

E09. Programming hours. Hours per term devoted to programming or model development

Answer (hours per term): _____

E10. Student AI projects. Number of student AI projects (coursework, theses, capstones) completed in the past academic year

Answer (projects): _____

E11. Undergraduate AI courses. Name the undergraduate course(s) in this department that are dedicated to AI or contain a dedicated AI module

#	Course code and title	AI hours per term	Status
1			
2			
3			

Status codes: 1 = Elective; 2 = Required

E12. Graduate / residency AI courses. Name the graduate or residency course(s) in this department that include AI content

#	Course code and title	AI hours per term	Status
1			
2			
3			

Status codes: 1 = Elective; 2 = Required

E13. Graduate AI offering. Graduate-level AI offering in this department *[domain CUR]*

□ (1) No graduate AI course
□ (2) Elective only
□ (3) Required course exists
□ (4) Required course plus a formal AI research project requirement
□ (-1) Don't know / Not applicable

E14. AI theses (3 y). Number of master's/doctoral theses or residency projects on AI defended in this department in the past 3 years

Answer (theses): _____

E15. Clinical AI training. Are students trained in clinical application and critical interpretation of AI outputs (reading AI-assisted reports, identifying AI errors, understanding limitations)? *[domain CUR]*

□ (1) None
□ (2) Informal discussion only
□ (3) 1 dedicated session
□ (4) Integrated into clinical rotations with formal assessment
□ (-1) Don't know / Not applicable

E16. Assessment of AI competency. How is student AI competency assessed? *[domain CUR]*

- ☐ (1) Not assessed
- ☐ (2) Written exam only
- ☐ (3) Separate AI assignment or project
- ☐ (4) Competency-based assessment (OSCE station, portfolio, practical demonstration)
- ☐ (-1) Don't know / Not applicable

E17. Teaching methods. Teaching methods used for AI content

- ☐ Lectures
- ☐ Flipped classroom
- ☐ Case-based learning
- ☐ Problem-based learning
- ☐ Journal club / critical appraisal
- ☐ Computer lab / coding exercises
- ☐ Hackathon / competition
- ☐ Supervised clinical use of AI tools
- ☐ Simulation with AI feedback
- ☐ AI tutor / chatbot-assisted learning

Other, specify: _____

E18. Educational materials. Dedicated AI educational materials (textbooks, online modules, case libraries) *[domain CUR]*

- ☐ (1) None used
- ☐ (2) Scattered resources, not formally organized
- ☐ (3) Some dedicated materials used
- ☐ (4) Comprehensive, curated collection integrated into the course
- ☐ (-1) Don't know / Not applicable

E19. Materials used. Name the main AI educational materials used

#	Material (title / URL)
1	
2	
3	

E20. Competency framework. Is AI teaching aligned with a competency framework (e.g., IADR/ITU-WHO core curriculum for dental AI, national curriculum guidelines)? *[domain CUR]*

Country adaptation: see table "Country context", row "Curriculum / accreditation body".

- ☐ (1) No framework
- ☐ (2) Aware of frameworks, not used
- ☐ (3) Partially aligned
- ☐ (4) Explicitly mapped to a framework with defined learning outcomes
- ☐ (-1) Don't know / Not applicable

E21. AI as a teaching tool. Is AI used as a tool to deliver teaching (adaptive learning, automated feedback on preparations, AI-generated cases, virtual patients)? *[domain CUR]*

- ☐ (1) Not used
- ☐ (2) Experimented by individual faculty
- ☐ (3) Used regularly in some activities
- ☐ (4) Systematically used and evaluated
- ☐ (-1) Don't know / Not applicable

E22. CPD AI offerings. Number of continuing-education (CPD) activities on AI offered by this department in the past 12 months

Answer (activities): _____

E23. Student preparedness. How prepared and interested in AI are students in this course? *[domain CUR]*

- ☐ (1) Low interest and preparedness
- ☐ (2) Interest but low preparedness
- ☐ (3) Interest and moderate preparedness
- ☐ (4) High interest and preparedness
- ☐ (-1) Don't know / Not applicable

E24. Planned curriculum change. Is an increase of AI content planned for the next 2 years? *[domain CUR]*

- ☐ (1) No plans
- ☐ (2) Under informal discussion
- ☐ (3) Approved, not yet implemented
- ☐ (4) Being implemented now
- ☐ (-1) Don't know / Not applicable

Part F. Generative AI in teaching and learning

Generative AI includes chatbots and large language models such as ChatGPT, Gemini, Copilot, DeepSeek, Qwen or Ernie Bot.

F01. Generative AI policy. Which policy governs student and faculty use of generative AI in this department/course? *[domain GEN]*

- ☐ (1) No policy
- ☐ (2) General academic-integrity policy only, not AI-specific
- ☐ (3) Interim guidance issued
- ☐ (4) Formal, regularly updated policy with training
- ☐ (-1) Don't know / Not applicable

F02. Policy name. Name of the policy document (and level: university / school / department / course)

Answer: _____

F03. Permitted student uses. Which student uses of generative AI are permitted in this course?

- ☐ Studying / explanations
- ☐ Writing support with disclosure
- ☐ Practising clinical cases
- ☐ Coding assistance
- ☐ Literature search / research support
- ☐ Use during assessments

- ☐ All use is prohibited
- ☐ Not defined

Other, specify: _____

F04. Prompt design teaching. Are students taught how to design effective prompts (context, task, format, audience, constraints)? *[domain GEN]*

- ☐ (1) Not taught
- ☐ (2) Mentioned informally
- ☐ (3) 1 dedicated session
- ☐ (4) Integrated practice with feedback
- ☐ (-1) Don't know / Not applicable

F05. Verification of outputs. Are students taught to verify generative-AI outputs (hallucinations, fabricated references, outdated guidance)? *[domain GEN]*

- ☐ (1) Not taught
- ☐ (2) Mentioned informally
- ☐ (3) 1 dedicated session
- ☐ (4) Integrated practice with assessment
- ☐ (-1) Don't know / Not applicable

F06. Generative AI safeguards. Which safeguards are in place for generative AI use in this department/course? (adapted from PAHO AI-GUARD, Tier 1)

Item	Yes	Partial	No	Don't know
Rule that no identifiable patient data is entered into public AI tools	☐	☐	☐	☐
An institutionally licensed, privacy-protected generative AI tool is available	☐	☐	☐	☐
Students must disclose AI use in assignments	☐	☐	☐	☐
Human verification required before AI output is used in patient care	☐	☐	☐	☐
Process to report misuse or AI errors	☐	☐	☐	☐
Staff and students trained on limitations and bias of generative AI	☐	☐	☐	☐

Scoring: Yes = 2, Partial = 1, No = 0.

F07. Faculty GenAI use. How often do faculty use generative AI to prepare teaching materials or assessments?

- ☐ Never
- ☐ Occasionally
- ☐ Monthly
- ☐ Weekly or more
- ☐ Don't know

F08. AI-detection software. Is AI-text detection software used to assess student work?

- ☐ Yes, routinely
- ☐ Yes, occasionally
- ☐ No
- ☐ Don't know

Part G. Governance, ethics, bias and regulation

Items G01–G07 refer to the named department/course; G08–G12 refer to the policies and procedures that govern it.

G01. Written AI plan. Does this department/course have a written plan or formal strategic direction for AI integration in teaching? *[domain GOV]*

- ☐ (1) None
- ☐ (2) Informal consensus, not documented
- ☐ (3) Draft document under discussion
- ☐ (4) Formal strategic plan, regularly reviewed
- ☐ (-1) Don't know / Not applicable

G02. Leadership commitment. How committed is the leadership of this department (head, division chief) to advancing AI initiatives? *[domain GOV]*

- ☐ (1) No commitment shown
- ☐ (2) Initial interest, no concrete actions
- ☐ (3) Some commitment, inconsistent or limited in scope
- ☐ (4) Strong commitment with ongoing initiatives and resources
- ☐ (-1) Don't know / Not applicable

G03. Oversight structure. Is there an AI committee or a named person accountable for AI in teaching and clinics? *[domain GOV]*

- ☐ (1) None
- ☐ (2) Informal responsibility
- ☐ (3) Committee or person named, limited mandate
- ☐ (4) Formal committee with mandate, reporting line and regular meetings
- ☐ (-1) Don't know / Not applicable

G04. AI ethics teaching. Does this course teach AI ethics (bias, privacy, informed consent, accountability, equity)? *[domain GOV]*

- ☐ (1) None
- ☐ (2) Brief mention only
- ☐ (3) 1 dedicated session
- ☐ (4) Systematic integration with case-based discussion and assessment
- ☐ (-1) Don't know / Not applicable

G05. Ethics topics. Which AI ethics topics are covered?

- ☐ Algorithmic bias and equity
- ☐ Privacy and data protection
- ☐ Informed consent for AI use with patients
- ☐ Professional accountability / liability
- ☐ Transparency and explainability
- ☐ Automation bias / over-reliance
- ☐ Intellectual property and authorship
- ☐ Environmental impact of AI
- ☐ Academic integrity
- ☐ None

G06. Bias teaching. Are students taught to check AI performance across patient subgroups and to recognize biased training data (PAHO Bias-free AI)? *[domain GOV]*

- □ (1) Not taught
- □ (2) Mentioned informally
- □ (3) 1 dedicated session
- □ (4) Hands-on exercises with real or simulated subgroup analyses
- □ (-1) Don't know / Not applicable

G07. Ethical guidelines available. Are established ethical guidelines or white papers on AI in dental education or practice available to faculty and students? *[domain GOV]*

- □ (1) None available
- □ (2) Some documents exist but are incomplete
- □ (3) Guidelines exist but need dissemination
- □ (4) Comprehensive, well-implemented guidelines available
- □ (-1) Don't know / Not applicable

G08. Guidance documents used. Which of the following documents are used in teaching or clinical protocols of this department/course?

- □ WHO — Ethics and governance of AI for health (2021) and guidance on large multi-modal models (2024/2025)
- □ UNESCO — Guidance for generative AI in education and research (2023)
- □ IADR / ITU-WHO Focus Group — Core curriculum for oral and dental AI (Schwendicke et al., 2023)
- □ AI reporting guidelines (CONSORT-AI, SPIRIT-AI, TRIPOD+AI, STARD-AI, CLAIM, dental AI checklist)
- □ PAHO/WHO AI toolkits (Readiness Assessment, AI-GUARD, Bias-free AI, Prompt design) [CA, US, BR]
- □ ADA Technical Report 1109, ANSI/ADA Standard 1110-1:2025, ADA White Paper 1106 [US, CA]
- □ CODA accreditation standards (AI/technology-related elements) [US]
- □ HIPAA privacy and security rules applied to AI tools [US]
- □ FDA guidance on AI-enabled device software functions [US]
- □ ADEA guidance or competencies relating to AI [US, CA]
- □ Canadian Dental Association position statements / guidance on AI [CA]
- □ CDAC accreditation standards (technology-related elements) [CA]
- □ PIPEDA / provincial health-privacy laws (PHIPA, Quebec Law 25) [CA]
- □ Health Canada guidance for machine-learning-enabled medical devices [CA]
- □ CFO resolutions / guidance on digital dentistry and AI [BR]
- □ LGPD (Lei 13.709/2018) — health data protection [BR]
- □ ANVISA RDC 657/2022 — Software as a Medical Device [BR]
- □ DCN Odontologia (Resolução CNE/CES 3/2021) [BR]
- □ PL 2338/2023 — Brazilian AI regulatory bill [BR]
- □ CNS/CONEP research-ethics resolutions (466/2012, 510/2016) [BR]
- □ Personal Information Protection Law (2021) / Data Security Law (2021) [CN]
- □ NMPA guidelines for registration review of AI medical device software [CN]
- □ Interim Measures for the Management of Generative AI Services (2023) [CN]
- □ National Health Commission guidance on AI application scenarios in health care [CN]
- □ Chinese Stomatological Association expert consensus / guidance on AI [CN]

- ☐ Ministry of Education guidance on AI in higher education [CN]
- ☐ Dental Council of India regulations / National Dental Commission Act 2023 [IN]
- ☐ ICMR Ethical Guidelines for Application of AI in Biomedical Research and Healthcare (2023) [IN]
- ☐ Digital Personal Data Protection Act 2023 and Rules [IN]
- ☐ CDSCO Medical Devices Rules 2017 — software as a medical device [IN]
- ☐ Ayushman Bharat Digital Mission (ABDM) health-data standards [IN]
- ☐ Indian Dental Association guidance on AI [IN]
- ☐ None of these

G09. Regulatory guidance incorporated. Is national / professional regulatory guidance on AI in dentistry incorporated into teaching and clinical protocols? *[domain GOV]*

Country adaptation: see table "Country context", row "Professional and regulatory bodies".

- ☐ (1) Not incorporated
- ☐ (2) Aware but not incorporated
- ☐ (3) Discussed in faculty meetings
- ☐ (4) Incorporated into teaching and clinical protocols
- ☐ (-1) Don't know / Not applicable

G10. Patient consent for AI. Are patients informed / asked for consent when AI is used in their care in the school clinic? *[domain GOV]*

- ☐ (1) Never
- ☐ (2) Occasionally, at clinician discretion
- ☐ (3) Required by policy but inconsistently applied
- ☐ (4) Required by policy, documented in the record
- ☐ (-1) Don't know / Not applicable

G11. AI tool governance (AI-GUARD). Before and during use of clinical AI tools in this department/course, are the following in place? (adapted from PAHO AI-GUARD, Tier 2)

Item	Yes	Partial	No	Don't know
Defined owner / responsible authority for each AI tool	☐	☐	☐	☐
Vendor documentation or model card reviewed (intended use, training data, limitations)	☐	☐	☐	☐
Independent / external validation evidence reviewed	☐	☐	☐	☐
Regulatory clearance verified for the intended use	☐	☐	☐	☐
Performance across patient subgroups evaluated	☐	☐	☐	☐
Human review required before acting on AI output	☐	☐	☐	☐
Clinician override possible and logged	☐	☐	☐	☐
Incident / error reporting process defined	☐	☐	☐	☐
Update / version-change notification procedure defined	☐	☐	☐	☐
Ongoing performance monitoring (drift) planned	☐	☐	☐	☐

Scoring: Yes = 2, Partial = 1, No = 0.

G12. Stakeholder involvement. How involved are stakeholders (faculty, students, clinical staff, IT, patients, external partners) in decisions about AI integration? *[domain GOV]*

- ☐ (1) Not involved

- ☐ (2) Limited, non-systematic involvement
- ☐ (3) Involvement exists but inconsistent
- ☐ (4) Well-established processes for systematic stakeholder input
- ☐ (-1) Don't know / Not applicable

Part H. Implementation, monitoring, evaluation and sustainability

Refers to AI initiatives of the named department/course.

H01. Pilot projects. Name any AI pilot project currently running in this department/course

#	Project	Area
1		
2		
3		

Area codes: 1 = Teaching / learning; 2 = Assessment; 3 = Clinical care; 4 = Research; 5 = Administration

H02. Pilot maturity. Status of AI pilot projects *[domain IMP]*

- ☐ (1) No pilot projects
- ☐ (2) Initial pilots with limited outcomes
- ☐ (3) Pilots ongoing, outcomes not yet assessed
- ☐ (4) Pilots with positive, documented outcomes
- ☐ (-1) Don't know / Not applicable

H03. Scalability. Potential for scaling successful AI initiatives to other courses *[domain IMP]*

- ☐ (1) No potential
- ☐ (2) Limited potential; substantial effort required
- ☐ (3) Some potential, requires significant resources and coordination
- ☐ (4) High potential with a clear pathway
- ☐ (-1) Don't know / Not applicable

H04. Workflow integration. How well are AI tools integrated into the clinical and teaching workflows of this department? *[domain IMP]*

- ☐ (1) Not integrated
- ☐ (2) Used outside routine workflow
- ☐ (3) Partly integrated
- ☐ (4) Fully integrated in routine workflow
- ☐ (-1) Don't know / Not applicable

H05. Performance metrics. Are there established metrics for monitoring the performance or impact of AI teaching initiatives? *[domain IMP]*

- ☐ (1) No metrics
- ☐ (2) Limited metrics
- ☐ (3) Metrics exist but need refinement
- ☐ (4) Comprehensive metrics with regular review
- ☐ (-1) Don't know / Not applicable

H06. Evaluation framework. Is the impact of AI teaching evaluated with a framework (e.g., Kirkpatrick levels: reaction, learning, behaviour, results)? *[domain IMP]*

- ☐ (1) Not evaluated
- ☐ (2) Student satisfaction only
- ☐ (3) Learning outcomes measured
- ☐ (4) Learning and clinical/behavioural outcomes measured
- ☐ (-1) Don't know / Not applicable

H07. Feedback mechanisms. Are there mechanisms to collect and act on feedback from students and faculty about AI initiatives? *[domain IMP]*

- ☐ (1) None
- ☐ (2) Informal feedback only
- ☐ (3) Formal collection, inconsistent follow-up
- ☐ (4) Formal collection with documented actions
- ☐ (-1) Don't know / Not applicable

H08. Continuous improvement. Are there mechanisms for continuous improvement based on evaluation of AI initiatives? *[domain IMP]*

- ☐ (1) No mechanisms
- ☐ (2) Limited mechanisms
- ☐ (3) Mechanisms exist but need better integration
- ☐ (4) Comprehensive feedback loops
- ☐ (-1) Don't know / Not applicable

H09. Cost-benefit evaluation. Has the cost and benefit of AI tools/initiatives been evaluated (licences, hardware, staff time vs. outcomes)? *[domain IMP]*

- ☐ (1) Never
- ☐ (2) Informal estimate
- ☐ (3) Formal evaluation for some initiatives
- ☐ (4) Formal evaluation for all major initiatives
- ☐ (-1) Don't know / Not applicable

H10. Financial sustainability. Are there plans for long-term financial sustainability of AI initiatives (budget lines, grant strategies, partnerships)? *[domain IMP]*

- ☐ (1) No plans
- ☐ (2) Initial discussions only
- ☐ (3) Plans exist but incomplete
- ☐ (4) Comprehensive, well-developed plans
- ☐ (-1) Don't know / Not applicable

H11. Funding sources. Name each funding source that supports AI teaching or research in this department/course

#	Funding source	Type
1		
2		
3		

Type codes: 1 = Institutional budget; 2 = National research agency (e.g., NIH, CIHR, CNPq/CAPES/FAPs, NSFC, ICMR/DST); 3 = Industry; 4 = Philanthropy / foundation; 5 = International agency; 6 = Other

H12. Funding adequacy. Funding available for AI teaching or research in this department/course *[domain IMP]*

- ☐ (1) No funding
- ☐ (2) Limited funding with significant gaps
- ☐ (3) Funding available but inconsistent or insufficient
- ☐ (4) Sufficient, reliable funding
- ☐ (-1) Don't know / Not applicable

H13. Annual AI budget. Approximate annual budget dedicated to AI in this department/course (USD equivalent)

- ☐ None
- ☐ Less than 5,000
- ☐ 5,000 – 25,000
- ☐ 25,001 – 100,000
- ☐ More than 100,000
- ☐ Don't know

H14. AI grants (3 y). Number of competitive AI-related grants obtained by this department in the past 3 years

Answer (grants): _____

Part I. Self-assessment and open response

I01. Overall maturity. Rate the overall AI integration maturity of this department/course

- ☐ (1) Exploring: no formal action; individual interest only
- ☐ (2) Planning: discussions underway, some resources identified, no systematic implementation
- ☐ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- ☐ (4) Mature: systematic curriculum integration, dedicated resources, continuous evaluation
- ☐ (-1) Don't know / Not applicable

I02. Primary barriers. Primary barriers to AI integration in this department/course (select up to three)

Select up to 3.

- ☐ Lack of funding
- ☐ Lack of faculty expertise
- ☐ Lack of time in the curriculum
- ☐ Lack of leadership support
- ☐ Lack of clinical AI tools / infrastructure
- ☐ Ethical / legal uncertainty
- ☐ Lack of student interest or preparedness
- ☐ Lack of data access
- ☐ Accreditation / curriculum rules leave no room
- ☐ Language barriers (tools/materials not in local language)

Other, specify: _____

I03. Facilitators. Main facilitators of AI integration (select up to three)

Select up to 3.

- ☐ Faculty champions
- ☐ Leadership support
- ☐ Dedicated funding
- ☐ Interdisciplinary partners
- ☐ Industry partnerships
- ☐ Student demand
- ☐ Accreditation / national guidelines
- ☐ National or international networks

Other, specify: _____

I04. Most important need. What is the single most important resource or policy support this department/course needs to advance AI integration in the next 2 years?

I05. Innovative practice. Describe any innovative AI-related teaching practice implemented in this department/course (optional)

I06. Vision 2030. How do you expect AI to change teaching in this department/course by 2030? (optional)

I07. Confidence in answers. How confident are you in the accuracy of your answers?

- ☐ (1) Low
- ☐ (2) Moderate
- ☐ (3) High
- ☐ (4) Very high (checked with records)

I08. Follow-up interview. Would you agree to a short follow-up interview?

- ☐ Yes
- ☐ No

3. Scoring and consolidation

- **Course record.** Each form is stored as one record keyed by School × Department × Course code. Nothing is aggregated before storage.
- **Domain scores.** For each record, the mean of the 1–4 items of each domain is computed ("Don't know" = missing) and converted to a 0–100 index: (mean − 1) / 3 × 100. The overall index is the mean of the seven domain indices.
- **AI-GUARD and safeguard indices.** Items G11 and F06 are scored Yes = 2, Partial = 1, No = 0 and expressed as a percentage of the maximum over answered rows.
- **Hours.** AI hours per year = hours per term × terms per year (B05). Hours are reported by format (E02), by topic (E03) and for programming/model development (E09).
- **Department roll-up.** Scores are averaged across the courses of a department; the number of contributing courses (n) is reported.
- **School roll-up.** Department scores are averaged with weights equal to the number of courses each department contributed, so a department with five AI courses does not carry the same weight as one with a single course.
- **Country summary.** School scores are averaged per country; the number of schools is reported.
- **Transparency rule.** For each school, the list of named courses that produced the curriculum score (Part E) is published, so readers can see exactly which courses were assessed.

Data file

Answers are stored in a flat file with one row per department/course (access code) and one column per item — 361 answer columns in total. The codebook (docs/codebook.csv) lists every column, its label, type, domain and value codes. Personal contact data (respondent name, email, interviewer) are kept in a separate private file.

Methodological recommendations

- Translation: forward–backward translation into French (Quebec), Portuguese (Brazil) and Chinese, following WHO guidance, with reconciliation by bilingual dental educators. English is used in India and the rest of Canada and the USA.
- Content validity: expert panel (at least two experts per country) rating item relevance; item-level and scale-level content validity index (I-CVI ≥ 0.78, S-CVI/Ave ≥ 0.90), optionally with a two-round Delphi.
- Pilot: cognitive interviews with 2–3 department heads per country, then a pilot with about 10 departments to check timing and item clarity.
- Reliability: Cronbach's alpha / McDonald's omega per domain; test–retest on a 10% subsample after 2–4 weeks; agreement between two respondents of the same department where possible.
- Sampling frame: official lists of dental schools (ADEA and CODA in the USA, ACFD in Canada, MEC/e-MEC and ABENO in Brazil, MOE and the Chinese Stomatological Association in China, DCI/NDC in India). Stratify by country, public/private status and region; within each school, invite the departments that teach diagnostic imaging, restorative/prosthodontics, orthodontics, periodontics, oral surgery, public health and research methods, plus any department teaching a dedicated AI course.
- Ethics: approval from the coordinating research ethics board and local approval where required (e.g., CONEP in Brazil, institutional ethics committees in India and China). Respondents provide informed consent (item A14). Personal contact data are stored separately from answers.
- Reporting: STROBE for cross-sectional studies and CHERRIES for web-based surveys.
- Analysis: descriptive statistics by country and level; Kruskal–Wallis / Mann–Whitney tests for between-country and public/private comparisons; multilevel (mixed-effects) models with courses nested in departments and schools; heat maps of domain indices.

4. Related studies

Most published studies on AI in dental education measure perceptions, knowledge and attitudes of students or educators, often at one institution. Curriculum-document analyses exist for some countries (e.g., Brazilian master's programs). No multinational study has yet measured AI integration at the level of named departments and courses, combining infrastructure, human resources, instruction hours by topic, governance and sustainability. This study fills that gap and adds China and India, which are underrepresented in the dental-education AI literature relative to the size of their dental workforce.

Study	Setting / design	Key finding
PAHO/WHO & IDB, 2024	Readiness toolkit	Defines the readiness dimensions and the scoring/analysis steps adapted here.
PAHO/WHO & IDB, 2026 (AI-GUARD)	Governance tool	Tiered Yes/Partial/No governance checklist for AI tools; basis of items F06 and G11.
PAHO, 2026 (Bias-free AI)	Guidance	Representative data, subgroup performance reporting and human oversight; basis of items C16 and G06.
PAHO, 2025 (Prompt design)	Guidance	Prompt design as a core digital-literacy competency; basis of items F04–F05.
Schwendicke et al., 2023	Expert consensus (IADR, ITU/WHO FG-AI4H), Delphi	Core curriculum for dental AI in four domains, mostly at "knowledge" level; reference framework for item E20.
Uribe et al., 2024	Worldwide survey of 428 dental educators	31% already used AI chatbots, 64% saw potential; main concern was the lack of guidelines and training.
Kavadella et al., 2026	Multinational survey of dental students	Students want AI education (about 70%) but over half feel unprepared by current curricula.
El-Hakim et al., 2025	Scoping review, 17 studies	AI improves procedural accuracy and feedback but struggles with dexterity and judgment; VR often mislabelled as AI.
Ardabil, Iran, 2025	MAIRS survey, faculty vs students	Faculty readiness higher than students (65.9 vs 52.0 total; ≈3.0 vs 2.4 per item).
Dantas & da Silva, 2026	Curriculum analysis, 92 Brazilian master's programs	AI content in well under 1% of disciplines, concentrated in higher-rated programs.
Wu et al., 2026 (China)	343 students, Wuhan University	89.8% support AI in dental care, 84.2% want AI education; views differ by training stage.
Chongqing, 2026 (China)	Student survey	Use of AI tools and mobile learning apps among dental students.
North India, 2025	Faculty survey, orthodontics and other departments	Faculty see opportunities but report training gaps and ethical concerns.
Sheng et al., 2026 (USA)	243 faculty, residents, students; UTHealth Houston	66% of faculty and 73% of students use LLM tools; faculty report greater need for AI training.
Rifaat et al., 2026	333 students, 55 faculty; Saudi Arabia	Faculty reported far more AI knowledge than students (81.8% vs 27.9%); both want formal training.
Busch et al., 2024	4,596 medical, dental, veterinary students; 192 faculties, 48 countries	Positive attitudes and demand for more AI teaching; only 205 dental students.
Teng et al., 2022 (Canada)	Countrywide survey of health-care students	Students want AI in curricula; informs Canadian comparison.
Bisdas et al., 2021	3,133 medical and dental students, 63 countries	72% see AI as a partner rather than a competitor.

Suggested article titles

- Course-Level Mapping of Artificial Intelligence Readiness in Dental Education: A Multinational Assessment of Infrastructure, Human Resources, Curriculum Hours and Governance in Canada, the United States, Brazil, China and India

- Which Course, Which Department? A Specificity-Preserving Readiness Assessment of Artificial Intelligence Integration in Dental Schools of the Americas and Asia
- From Awareness to Implementation: Department-Level Maturity of Artificial Intelligence in Dental Education in Five Countries — A Cross-Sectional Study Using an Adapted PAHO/WHO Readiness Framework

Candidate journals

Journal of Dental Education (ADEA) · European Journal of Dental Education · BMC Medical Education · Journal of Dentistry · International Dental Journal (FDI) · JMIR Medical Education · Frontiers in Dental Medicine · Brazilian Oral Research.

References

1. Pan American Health Organization, Inter-American Development Bank. Artificial Intelligence in Public Health: Readiness Assessment Toolkit — Assessing and Enhancing Preparedness for AI Integration in Public Health. Washington, D.C.: PAHO; 2024.
2. Pan American Health Organization, Inter-American Development Bank. AI-GUARD Tool: Artificial Intelligence Governance & Use Assessment for Responsible Deployment. Washington, D.C.: PAHO; 2026.
3. Pan American Health Organization. Bias-free artificial intelligence in health: Dos and don'ts for developing and implementing algorithms. Washington, D.C.: PAHO; 2026. PAHO/EIH/IS/26-0001.
4. Pan American Health Organization. AI prompt design for public health: Using generative AI responsibly. Washington, D.C.: PAHO; 2025. PAHO/EIH/IS/25-0008.
5. Schwendicke F, Chaurasia A, Wiegand T, et al. Artificial intelligence for oral and dental healthcare: Core education curriculum. J Dent. 2023;128:104363. doi:10.1016/j.jdent.2022.104363
6. Uribe SE, Maldupa I, Kavadella A, et al. Artificial intelligence chatbots and large language models in dental education: Worldwide survey of educators. Eur J Dent Educ. 2024;28(4):865-876. doi:10.1111/eje.13009
7. Kavadella A, Uribe SE, Schwendicke F, et al. Dental students' knowledge, perceptions, and educational needs regarding artificial intelligence: a multinational cross-sectional survey. BMC Med Educ. 2026;26:401. doi:10.1186/s12909-026-08699-6
8. El-Hakim M, Anthonappa R, Fawzy A. Artificial Intelligence in Dental Education: A Scoping Review of Applications, Challenges, and Gaps. Dent J (Basel). 2025;13(9):384. doi:10.3390/dj13090384
9. Ahmadian Yazdi N, et al. Readiness to use artificial intelligence: a comparative study among dental faculty members and students. BMC Med Educ. 2025;25:1006. doi:10.1186/s12909-025-07621-w [verify author list]
10. Dantas JN, da Silva MAD. AI Integration in Brazilian Master's Curricula and Its Implications in Dental Education. Int J Dent. 2026;9974810. Preprint: Research Square rs-3395718. [verify final citation]
11. Wu Y, Bao Y, Jiang Z, et al. Preparing future dentists for artificial intelligence: a cross-sectional study of perceptions and educational needs in China. Front Public Health. 2026;14:1841849. doi:10.3389/fpubh.2026.1841849
12. A survey-based study on dental students' perceptions and attitudes toward AI and mobile medical learning apps in Chongqing, China. BMC Med Educ. 2026. doi:10.1186/s12909-026-10709-6 [verify authors]
13. Faculty perspectives on integrating artificial intelligence into orthodontic and interdisciplinary dental education: opportunities, challenges, and strategies. Cureus. 2025. PMCID: PMC12540188 [verify authors]
14. Sheng C, McFarland C, Angelov N, Eswaran SVK, Halpin R, Chang J. Evaluating artificial intelligence large language models in dental education: a cross-sectional survey on usage, perceptions, and integration at a U.S. dental school. Front Digit Health. 2026;8:1786363. doi:10.3389/fdgth.2026.1786363
15. Rifaat S, Gaffar B, Faridi MA, et al. Dental Students' and Educators' Perceptions Toward the Adoption of Artificial Intelligence in Dental Education: A Survey-Based Study. Int J Dent. 2026;2026:3313083. doi:10.1155/ijod/3313083
16. Busch F, Hoffmann L, Truhn D, et al. Global cross-sectional student survey on AI in medical, dental, and veterinary education and practice at 192 faculties. BMC Med Educ. 2024. PMID: 39342231
17. Teng M, Singla R, Yau O, et al. Health Care Students' Perspectives on Artificial Intelligence: Countrywide Survey in Canada. JMIR Med Educ. 2022;8(1):e33390.
18. Bisdas S, Topriceanu CC, Zakrzewska Z, et al. Artificial Intelligence in Medicine: A Multinational Multi-Center Survey on the Medical and Dental Students' Perception. Front Public Health. 2021;9:795284.

19. Karaca O, Çalışkan SA, Demir K. Medical artificial intelligence readiness scale for medical students (MAIRS-MS) – development, validity and reliability study. BMC Med Educ. 2021;21:112.

20. Thurzo A, Strunga M, Urban R, Surovková J, Afrashtehfar KI. Impact of Artificial Intelligence on Dental Education: A Review and Guide for Curriculum Update. Educ Sci. 2023;13(2):150.

21. American Dental Association. Technical Report No. 1109 (Image analysis systems using AI); ANSI/ADA Standard No. 1110-1:2025 (Validation dataset guidance, Part 1); White Paper No. 1106 (AI in dentistry). Chicago: ADA.

22. World Health Organization. Ethics and governance of artificial intelligence for health: WHO guidance. Geneva: WHO; 2021.